Moynuddin Al Masum

AI Engineer (Computer Vision)

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Social Info

LinkedIn
GitHub
LeetCode
Portfolio

Skills: Generative AI, Computer Vision, ML, RAG, LLMs, LangChain, n8n, FastAPI, NLP, TensorFlow, PyTorch, Scikit-learn, Neural Networks, Python, C++, Edge AI (DEEPX NPU), Docker, React

About Me

I work with Machine Learning, Deep Learning, and AI, using tools like TensorFlow, PyTorch, Scikit-learn, Python, C++, OpenCV, YOLOv8, ByteTrack, Keras, Pandas, and Matplotlib to build real-time, intelligent solutions. My projects include AI-based surveillance, fire & smoke detection, and automated attendance systems, integrating computer vision and NLP. As an AI Engineer (Computer Vision) at SmartData Technologies, I deploy real-time vision systems on the DEEPX DX-M1 NPU. I have hands-on experience in end-to-end model development, optimization, and deployment.

Education

Bangladesh University of Professionals
BSc in Information & Communication Engineering, Jun 2021 - Jun 2026

Experience

AI Engineer (Computer Vision) 1 Sep 2026 – Present
SmartData Technologies Limited

- Build and deploy real-time computer-vision systems on the DEEPX DX-M1 NPU — face-recognition attendance, hall management, conveyor bag counting and multi-tenant office analytics — using C++, GStreamer (VA-API GPU decoding) and Docker.
- Re-engineered the BUP face-attendance pipeline into a C++ NPU engine (rank-1 accuracy 92.8% → 99.0%, 1.7× throughput, 3.1× less CPU) and built the BUP Smart Hall, CCPLC bag-counting and Smart Office analytics platforms end to end.

AI Engineer Intern 1 May 2026 – 30 Aug 2026 (4 months)
SmartData Technologies Limited | Internship Certificate

- Completed a 4-month internship in Artificial Intelligence Engineering (computer vision), then continued as a full-time AI Engineer from 1 September 2026.

Projects

Automatic Attendance System for University: *Video*
Built a real-time classroom attendance system for BUP: a C++ engine (SCRFD + ArcFace) on the DX-M1 NPU marks students present from IP-CCTV; rank-1 accuracy rose from 92.8% to 99.0%.

- Tools Used: C++, pybind11, DEEPX DX-M1 NPU (DXRT), SCRFD, ArcFace, OpenCV, GStreamer, RTSP / IP-CCTV, Flask, Docker.

Smart Office – AI Office Analytics Platform: *Video*
Built a multi-tenant office-analytics platform running three models in one C++ binary on the DX-M1 NPU: attendance, restricted-zone, fall and desk modules with alerts; 2.6× throughput.

- Tools Used: C++17, DEEPX DX-M1 NPU (DXRT), YOLO26-pose, SCRFD, ArcFace, GStreamer, Drogon, PostgreSQL, React, Docker.

Smart Hall Management System: *Video*
Built a hall-management platform for BUP as one C++20 server in Docker: face-recognition gate and dining cameras on the DX-M1 NPU, meal planning and bills; 99.9% first-match accuracy.

- Tools Used: C++20, DEEPX DX-M1 NPU, SCRFD-10G, ArcFace, ByteTrack, GStreamer (VA-API), SQLite, RTSP, Docker.

CCPLC – AIoT Cement Bag Counting System: *Video*
Built a real-time bag-counting and loading-management system for conveyor belts: YOLOv8s on the DX-M1 NPU (~197 fps), operator-drawn count zone, shift reports and a React dashboard.

- Tools Used: YOLOv8, DEEPX DX-M1 NPU, C++17, OpenCV, GStreamer (VA-API), React, Vite, WebSocket / REST API, nginx, Docker.

Real-time Object Detection & Smart Counting System: *Link* | *Video*
Built a real-time IP-CCTV analytics system for object detection, multi-object tracking and bi-directional IN/OUT counting from RTSP streams, with stable ID tracking and optimized GPU performance.

- Tools Used: Ultralytics YOLOv8, Custom YOLO Model, OpenCV, CCTV RTSP Stream Integration, ByteTrack, TensorRT

BUP RAG Chatbot: *Link* | *Video*
Developed an intelligent FAQ chatbot for Bangladesh University of Professionals (BUP) using a Retrieval-Augmented Generation (RAG) pipeline that semantically searches FAQs with FAISS and generates accurate, language-adaptive answers via Google Gemini API in real time.

- Tools Used: FAISS, SentenceTransformer, Google Gemini API, Gradio, RAG (Retrieval-Augmented Generation)

AI LinkedIn Post Generator: *Link* | *Video*
Developed an AI-powered tool that generates professional, structured LinkedIn posts in multiple languages based on a given topic, using a user-friendly web interface.

- Tools Used: LangChain, OpenAI GPT-4o-mini, Gradio, NLP, Prompt Engineering

Dataset Contributions

Animal Detection Dataset, 15 Classes (Public, Roboflow Universe): *Link*
Annotated and openly published a 15-class animal detection dataset: 7,499 images, 14,996 bounding boxes and a fixed 70/20/10 split under CC BY 4.0, free for anyone to train and benchmark detectors.

Bangladeshi Vehicle Detection Dataset (Team of 3):
Collected and annotated 5,900+ Bangladeshi road images for 19 local vehicle classes (CNG, leguna, easybike, rickshaw, horse cart, launch); merged three separately labelled datasets into one Roboflow project with a review workflow and Gemini-assisted pre-labelling.

Publications (Under Review)

- Multi-Agent Framework for Pediatric Pneumonia Diagnosis Integrating Vision and Language Models (**Accepted**)
- A Hybrid Deep-Learning Framework with Interpretable Risk Scoring for Early Gestational Diabetes Prediction